

$$\underline{R(f)} = \int_a^b \underbrace{K_m(t)}_{\text{kernel}} \underbrace{f^{(m+1)}(t)}_{\text{kernel}} dt$$

degree of precision = m
 $d = m$

$$\nabla (x-t)_+ = \begin{cases} x-t & , x \geq t \\ 0 & , x \leq t \end{cases}$$

$$K_m(t) = \frac{1}{m!} R_m(\underline{(x-t)_+^m})$$

$$f(x) = L_m f(x) + R_m(f)$$

$$R_m(f) = f - L_m f$$

$$R_m(e_j)$$

$$\int f = \text{Aprox}(f) + R(f)$$

$$R(f) = \int f - \text{Aprox}(f)$$

$$R((x-t)_+^n) = \int (x-t)_+^n - \text{Aprox}((x-t)_+^n)$$

$$-2f'(-1) + 2f(1) + 2[(x-t)_+^n]'_{x=-1} + 2[(x-t)_+^n]_{x=1}$$

$$\det(A - \lambda I) = 0$$

$\hookrightarrow$ sol. λ = eigenvalues A

$$A \mathbf{v} = \lambda \mathbf{v}$$

$\uparrow$ eigenvector $\uparrow$ eigenvalue

Examples

- QR

- Cholesky

$(0, 1), (1, -3), (2, 7)$

Linear poly. = $\boxed{f(x) = ax + b}$

$$E(a, b) = \sum_{k=1}^3 [f(x_k) - p(x_k)]^2$$

$$= \sum_{k=1}^3 [f(x_k) - ax_k - b]^2$$

$$= (1 - a \cdot 0 - b)^2 + (-3 - a \cdot 1 - b)^2 + (7 - a \cdot 2 - b)^2$$

$$= (1 - b)^2 + (-3 - a - b)^2 + (7 - 2a - b)^2$$

$$\begin{cases} \frac{\partial E}{\partial a}(a, b) = 0 \\ \frac{\partial E}{\partial b}(a, b) = 0 \end{cases}$$

Hermite : $\begin{pmatrix} x_0 \\ \vdots \\ x_m \end{pmatrix} \begin{matrix} x_0 = 0 \\ x_1 = 1 \end{matrix} \quad , \quad \begin{matrix} f^{(0)}(x_0) = 1 \\ f^{(0)}(x_1) = 0 \end{matrix} \quad , \quad \begin{matrix} f^{(1)}(x_0) = 2 \\ \text{---} \end{matrix} \quad \left| \quad \begin{matrix} h_0 = 1 \\ h_1 = 0 \end{matrix} \right.$

x_k	$f(x_k)$
0	1
1	-3
2	7

$$H_m f(x) = \sum_{k=0}^m \sum_{j=0}^{l_k} h_{kj}(x) f^{(j)}(x_k)$$

l_k → $l_0 = 1$
 $l_1 = 0$

$$(m) = m + l_0 + \dots + l_{m-1}$$

degree of $H_m f$

$$m = 1$$

$$(m) = \sum_{k=0}^m (l_k + 1) - 1$$

$$= (l_0 + 1) + (l_1 + 1) - 1$$

$$= 1 + 1 + 0 + 1 - 1 = 2$$

$$\Rightarrow H_2 f(x) = \sum_{k=0}^2 \sum_{j=0}^{l_k} h_{kj}(x) f^{(j)}(x_k)$$

$$= \underbrace{h_{00}(x) f^{(0)}(x_0)}_{k=0, j=0} + \underbrace{h_{01}(x) f^{(1)}(x_0)}_{j=1} + \underbrace{h_{10}(x) f^{(0)}(x_1)}_{k=1, j=0}$$

$$= h_{00}(x) \cdot 1 + h_{01}(x) \cdot 2 + \cancel{h_{10}(x) \cdot 0}$$

Heute fundamental polynomials

$$h_{00}(x) = ax^2 + bx + c$$

$$h'_{00}(x) = 2ax + b$$

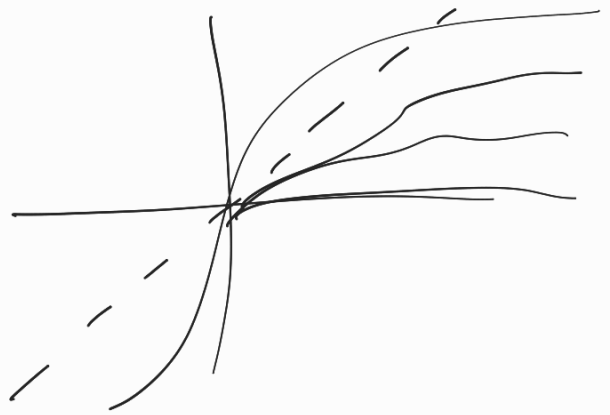
$$\begin{cases} h_{00}(x_0) = 1 \\ h'_{00}(x_0) = 0 \\ h_{00}(x_1) = 0 \end{cases} \Leftrightarrow \begin{cases} ax_0^2 + bx_0 + c = 1 \\ 2ax_0 + b = 0 \\ ax_1^2 + bx_1 + c = 0 \end{cases} \Leftrightarrow \begin{cases} c = 1 \\ b = 0 \\ a + b + c = 0 \end{cases}$$

$$\Rightarrow a = -1$$

$$\Rightarrow h_{00}(x) = -x^2 + 1$$

$$g(x) = \frac{1 - \sin x}{2}$$

$$g(x) = x \Leftrightarrow 1 - \sin x = 2x$$



$$g: [0, 1] \rightarrow \mathbb{R}, \quad g(x) = \frac{1 - \sin x}{2}$$

i. $g \in C^1[0, 1]$

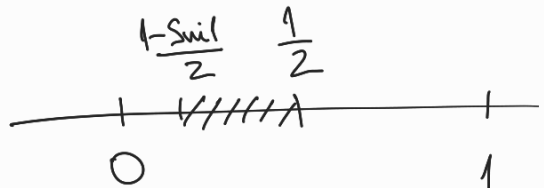
ii. $\forall x \in [0, 1]: g(x) \in [0, 1]$

$$g'(x) = \frac{-\cos x}{2}$$

x	0	1
$g(x)$	1/2	$\frac{1 - \sin 1}{2}$
$g'(x)$	-	-

$$\forall x \in [0, 1]:$$

$$g(x) \in \left[\frac{1 - \sin 1}{2}, \frac{1}{2} \right] \subseteq [0, 1]$$



iii. $\max_{x \in [0, 1]} |g'(x)| = \max_{x \in [0, 1]} \left| \frac{-\cos x}{2} \right| \leq \frac{1}{2} \cdot 1 = \frac{1}{2} < 1$

$$|\cos x| \leq 1$$

T.B.

$$\Rightarrow \exists! \alpha \in [0, 1] \text{ s.t. } g(\alpha) = \alpha$$

$$\alpha = \lim_{n \rightarrow \infty} x_n, \quad x_n = g(x_{n-1}), \quad \forall n \geq 1$$

$$x_0 \in [0, 1]$$

$$\begin{array}{lcl}
 x_0 = 0 & , & \overline{f(x_0)} \\
 x_1 = 1 & , & \overline{f'(x_1)} \\
 x_2 = 2 & , & \overline{f''(x_2)}
 \end{array}
 \quad \text{Birkhoff}$$

(Hermite lacunary)

$$\textcircled{x_0}: I_0 = \{0\}$$

$$\textcircled{x_1}: I_1 = \{1\}$$

$$\textcircled{x_2}: I_2 = \{0, 2\}$$

$$M = 2 \quad (\text{index of the last point})$$

$$m = |I_0| + |I_1| + |I_2| - 1$$

$\uparrow$ degree of $\uparrow$ and $(I_0) = \text{nb. of elements}$
 $\text{Bem } f$

$$B_m f(x) = \sum_{k=0}^m \sum_{j \in I_k} b_{kj}(x) f^{(j)}(x_k)$$

Here, $m = 1 + 1 + 2 - 1 = 3 \Rightarrow m = 3$

$$\Rightarrow B_3 f(x) = \sum_{k=0}^2 \sum_{j \in I_k} b_{kj}(x) f^{(j)}(x_k)$$

$$\begin{aligned}
 = & \underbrace{b_{00}(x)}_{\substack{k=0 \\ j \in I_0 = \{0\} \\ j=0}} f^{(0)}(x_0) + \underbrace{b_{11}(x)}_{\substack{k=1 \\ j \in I_1 = \{1\} \\ j=1}} f'(x_1) + \underbrace{b_{20}(x)}_{\substack{k=2 \\ j \in I_2 = \{0, 2\} \\ j=0}} f^{(0)}(x_2) + \underbrace{b_{22}(x)}_{\substack{k=2 \\ j \in I_2 = \{0, 2\} \\ j=2}} f''(x_2)
 \end{aligned}$$

Spline cubic
 $\left[\begin{array}{l} (2.9) + (2.10) \\ \text{Natural} \quad (2.13) \end{array} \right] \rightarrow m_k$
 $\downarrow$
 $(2.4) + (2.5)$

